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MIL-P-15736/9(SHIPS)
8 September 1970

MILITARY SPECIFICATION

POWER SUPPLY, CATHODIC PROTECTION, AUTOMATIC HULL POTENTIAL TYPE, INACTIVE FLEET

1. SCOPE

1.1 This specification covers the requirements for automatically controlled rectifiers of the constant hull potential type for the cathodic protection of inactive ships. The direct current (d.c.) output of the rectifier is automatically controlled by a saturable reactor to maintain a constant hull polarization potential which compensates for variations in environmental parameters.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-250 - Aluminum and Aluminum Alloy Plate and Sheet, General Specification for.
QQ-S-775 - Steel Sheets, Carbon, Zinc-Coated.

MILITARY

MIL-M-10304 - Meters, Electrical Indicating, Panel Type, Ruggedized, General Specification for.
MIL-P-15736 - Power Supplies, Metallic Rectifier, Naval Shipboard.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Power supplies shall conform to the requirements of class R power supplies of MIL-P-15736 except that qualification does not apply and as specified hereinafter. (Where any conflict exists between MIL-P-15736 and this specification, the latter shall govern.)

3.2 Electrical ratings. Electrical ratings shall be as specified in table I.

Table I - Electrical ratings.

Parameter	Requirements
Input power:	
Alternating current (a.c.)	440 volts
Frequency (input)	60 hertz
Phase (input)	Single phase
Average d.c. volts (output)	15 volt capability
Average d.c. amperes (output)	250 amp capability
Duty	Continuous
Type of load	Cathodic protection (resistive load)
D.c. output ammeter	0-300 amps
D.c. output voltmeter	0-30 volts
D.c. hull polarization potential voltmeter	-1.0 volt, 0 center, +1.0 volt

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3.3 A.c. input.

3.3.1 Rectifiers. Rectification of the output of the power transformer shall be accomplished by means of a combination of one or more rectifier stacks or assemblies in a full wave arrangement. The use of combinations of standard size rectifying devices is encouraged, so that, for a given manufacturer, the number of different kinds of rectifying devices used shall be as small as possible.

3.3.2 Rectifying agent. The rectifying agent shall be silicon diodes.

3.4 D.c. output. The d.c. output voltage of the power supply shall be controlled by a saturable reactor so that the pre-set polarization potential is maintained.

3.4.1 Output control. The automatic control shall be provided by a saturable reactor and control assembly which continuously monitors the hull polarization potential with respect to the silver-silver chloride reference electrode (see 3.11.2) and varies the saturable reactor output to maintain the hull polarization potential within plus or minus 0.05 volts of the value set on the control rheostat. The saturable reactor, shall provide continuous control of the rectifier output over a range of 5 to 100 percent of the rated output (see 3.2). The equipment shall maintain the hull polarization potential at any pre-set level between minus 0.70 to minus 1.10 volts with respect to a silver-silver chloride reference electrode. When the hull polarization potential is within plus or minus 0.05 volts of the pre-set potential, the current flow between the reference electrode and the power supply shall not exceed 20 microamperes. A separate signal ground provision shall be provided.

3.5 Electrical connections. The input and output electrical connections shall be brass stud type terminals. The output terminals shall be accessible without opening the enclosure, but shall be protected against weather and inadvertent contact by personnel. Minimum terminal size shall be determined by the maximum current through the connection.

3.5.1 Input. A terminal board shall be provided with all terminals clearly marked showing connection points for 440 volt input, reference cell signal, and signal ground.

3.5.2 Output. Power supplies shall be provided with terminals for convenient connection of output load leads. At least 6 ship ground (negative) terminals on a common bus and 6 anode (positive) terminals shall be provided. To accommodate varying cable lengths, voltage drops and hull bottom conditions for multiple ship arrangements, the anode terminals will be divided into 3 groups of 2 terminals which simultaneously provide output levels at 100, 66, and 33 percent of the output voltage of the saturable reactor. Each level must be capable of carrying the full current output.

3.6 Overload protection. The saturable reactor control circuit shall be such that overloading by a short circuit across the output will not interrupt the output current, but will limit the current to less than 120 percent of the full rated value. The rectifier shall be capable of operating indefinitely without harm at a short circuit condition. No fuses shall be used. Circuit breakers shall be permitted only at the a.c. power input. When the reference electrode is removed from the water, the output of the power supply shall be less than 10 percent of the rated output.

3.7 Ambient temperature. The operating ambient temperature range shall be minus 20 to plus 50°C.

3.8 Ventilation. Ventilation shall be natural draft.

3.9 Enclosure. The enclosure shall be dripproof protected with a hinged front access door in accordance with MIL-P-15736. The "on-off" switch, control rheostat and meters shall be flush mounted on a rigid front member of the enclosure in an area protected from mechanical damage. All openings in the enclosure, other than the door, shall be screened to prevent the entrance of insects. The enclosure shall be fabricated from steel sheet conforming to QQ-S-775 or aluminum sheet conforming to QQ-A-250.

3.10 Size and weight. The maximum size shall be 24 cubic feet and the maximum weight shall be 650 pounds.

3.11 Components. All electronic components in the control circuits, except for those specified in 3.11.1 through 3.11.3 shall be magnetic amplifier or solid state devices.

3.11.1 Meters. Separate meters shall be provided for indicating the output voltage and load current of the power supply. A polarization meter shall be provided to indicate the variation of the actual hull polarization potential from the potential set on the

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control rheostat. A zero center scale indication will indicate a hull polarization potential equal to the set potential. The plus or minus 0.05 volt tolerance allowed on the set potential shall be marked on the scale in a different color from potential differences larger than that tolerance. Meters shall be 3-1/2 inches round panel type in accordance with MIL-M-10304. The meter full scale range shall be as indicated in table I. Meters shall be hermetically sealed, and shall be provided with an external "zero adjust" device. The meters shall be mounted on the power supply enclosure exterior so that they may be read without opening the enclosure door.

3.11.2 Silver-silver chloride reference electrode. The reference electrode shall be of the silver-silver chloride type with at least 10 square inches of silver mesh as the active surface. The electrode shall be protected from mechanical damage by a non-metallic encasement. A 75 foot, 14 gauge or larger water-tight cable lead shall be provided with the electrode. The seal between the cable lead and the electrode encasement shall be water-tight. The electrode shall be designed for at least 5 years operation in salt water, and shall be accurate to plus or minus 0.01 volts relative to a laboratory standard silver-silver chloride electrode when tested as in 4.3.2.

3.11.3 Miscellaneous controls. The following miscellaneous controls shall be provided:

(a) A locking knob type rheostat connected to the control assembly to permit a smooth continuous adjustment of the hull polarization potential, as required, over the range of minus 0.70 to minus 1.10 volts.
(b) A switch shall be provided for on-off operation of the power supply at the a.c. input line. The switch shall disconnect both sides of the a.c. input.
(c) A rotary selector switch to permit the d.c. output voltmeter to measure the voltage on any of the three output voltage levels.
(d) Surge suppressors shall be provided on the a.c. input and the d.c. output to provide protection from lightning.

3.12 Repair parts. The following repair parts shall be supplied for each power supply in accordance with MIL-P-15736.

(a) One silicon diode of each type used for power rectification.
(b) One surge suppressor of each type used.

3.13 Drawings. Drawings shall be in accordance with MIL-P-15736.

3.13.1 A permanent laminated plastic schematic indicating the wiring of electrical components shall be attached to the inside of the enclosure door.

3.14 Technical manuals. Technical manuals shall be in accordance with MIL-P-15736.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance provisions shall be in accordance with MIL-P-15736, except for qualification which is not applicable to this equipment and the tests specified in 4.3 herein. Prior to government acceptance of the product, the manufacturer shall submit to the procuring activity the master drawings for the product, other specific information requested by the procuring activity, and test evidence that the product conforms to the requirements specified for the reference electrode in 3.11.2 herein, and conforms to all requirements of MIL-P-15736 including environmental requirements, except for shock and vibration. Environmental tests are required only once on any specific design of equipment. Any change in the product will require complete testing against all requirements. Selective and periodic inspections shall include tests of the reference electrode in addition to the tests specified in MIL-P-15736.

4.3 Tests.

4.3.1 Voltage regulation. The power supply shall be tested for steady state voltage regulation performance in accordance with the procedures of MIL-P-15736, except as modified in 4.3.1.1 herein.

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4.3.1.1 Steady state voltage regulation shall be measured at 4 equally spaced steps of output current and at resistive loads requiring from 5 percent of rated voltage to full rated load in approximately 5 equally spaced steps and overloaded to a short circuit condition.

4.3.2 Silver-silver chloride reference electrode. The silver-silver chloride reference electrode shall be tested by comparison with a laboratory standard silver-silver chloride reference electrode (see 3.11.2). The electrodes shall be tested in a laboratory scale cathodic protection system as shown on figure 1. The polarization potential measured by the silver-silver chloride reference electrode shall be compared to the potential measured by the laboratory standard electrode under the following conditions:

(a) No cathodic protection.
(b) Cathodic protection applied so that laboratory standard electrode measures a potential of approximately minus 0.85 volts.
(c) Same as condition (b) except voltage shall be minus 1.00 volts.

The reference electrode shall be immersed and electrically connected to the test circuit for 72 hours for each of the above conditions. The potential measurements shall be made with an instrument which draws no more current than specified for the control circuit. The laboratory standard electrode shall be immersed in the test tank and electrically substituted for the reference electrode at the beginning of the test period and every 8 hours thereafter, until the end of the test period.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-P-15736.

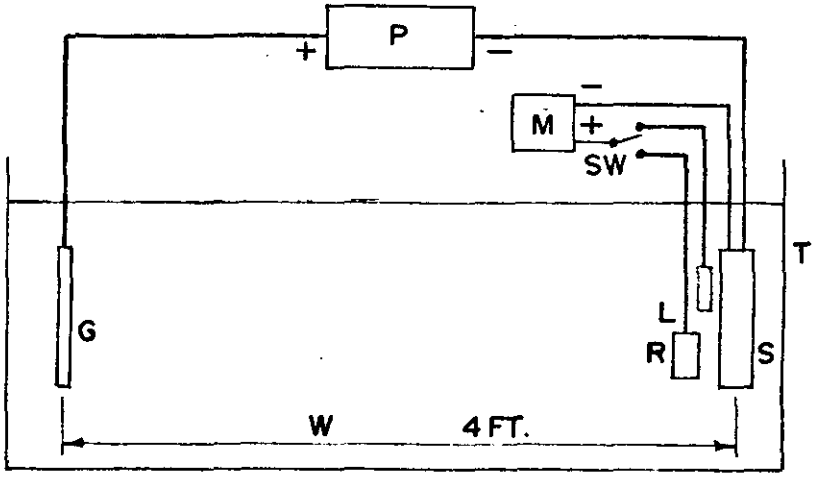
6. NOTES

6.1 The notes specified in MIL-P-15736 are applicable to this specification.

6.2 The silver-silver chloride reference electrode can be supplied as a separate item to the requirements of 3.11.2 of this specification.

Preparing activity:
Navy - SH
(Project 6130-N087)

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SYM	PART
G	GRAPHITE ANODE
L	LABORATORY STANDARD AG/AGCL ELECTRODE
M	VOLTMETER
P	LABORATORY POWER SUPPLY
R	REFERENCE ELECTRODE
S	STEEL SHEET (4 sq ft min.)
SW	SELECTOR SWITCH
T	TEST TANK
W	3 PERCENT SODIUM CLORIDE SOLUTION

SH9989

FIGURE 1 SCHEMATIC DIAGRAM OF REFERENCE ELECTRODE TEST

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